HARIKRISHNAN S

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OBJECTIVE

Motivated final-year CS student with a strong foundation in cybersecurity, IoT-based systems, and frontend/UI design. Experienced in building secure data-sanitization tools, IoT hardware (ESP32, sensor integration, Fusion 360 design) with real-time dashboards, and clean UI/branding using Figma. Passionate about turning ideas into practical, scalable solutions.

EDUCATION

B.E. Computer Science and Engineering
Akshaya College of Engineering and Technology — 7.27 CGPA (2023–2027)
HSC
Sri Sai Matriculation Higher Secondary School, Tirupur — 71% (2021–2023)
SSLC
Sri Sai Matriculation Higher Secondary School, Tirupur — 70.6% (2020–2021)

SKILL

- Programming: C, Python
- Web Development: HTML, CSS, React js
- Cybersecurity: Nmap, Burp Suite, Wireshark, Penetration Testing (Hack The Box, TryHackMe), Data Recovery Tools, CyberTools, Cybersecurity with IOT, Zeek and Wazuh
- Database: MongoDB, MySQL , Supabase
- Operating Systems: Linux, Ubuntu, Windows
- Tools: Git, GitHub
- Soft Skills: Problem Solving, Quick Learning, Teamwork, Public Speaking

INTERNSHIP

Cybersecurity & Ethical Hacking, Hack up Technology, Coimbatore (23 June 2025 - 23 July 2025 , 18 December 2025 - 18 January 2026 , 17 June 2026 - 17 July 2026)

- Completed a one-month intensive internship focused on cybersecurity and ethical hacking.
- Performed network scanning and vulnerability assessment using Nmap.
- Practiced penetration testing on Hack The Box and TryHackMe platforms.
- Worked with data recovery tools , cybersecurity tools and techniques.
- Worked in **“ IOT Project - SOS Emergency Response System”**
- Conducted web security testing using Burp Suite and network analysis using Wireshark.
- Developed skills in identifying, analyzing, and mitigating security threats.

PROJECT

CyberResilience AI: Intelligent Vulnerability Assessment and Network Intrusion Detection Platform

- Developing an AI-driven cybersecurity platform to analyse network telemetry, identify suspicious activity, assess security risks, and forecast potential attack progression.
- Designing a temporal ML pipeline that transforms PCAP/network-flow data into time-windowed network states for sequential modelling and multi-step attack-state forecasting using LSTM/GRU architectures.
- Integrating Scapy-based packet feature extraction, temporal aggregation, and MITRE ATT&CK technique mapping to contextualise observed security events.
- Building a React/Vite frontend and FastAPI backend for network asset discovery, website vulnerability assessment, security-event monitoring, incident management, and risk reporting.

- Integrating security assessment tools including Nmap, Nikto, Wapiti, SQLMap, and Gobuster to collect scan evidence and support vulnerability analysis.
- Planning explainable AI capabilities using SHAP and/or attention-based methods to interpret model predictions, alongside risk visualisation and automated security reports.
- Designing a SHA-256-based evidence-integrity mechanism with blockchain-inspired verification for tamper-evident incident records.
- Planning integration of network monitoring and detection tools such as Suricata, Zeek, and Wireshark/TShark to support traffic analysis, alert correlation, and anomaly detection.

Datanix – AI-Assisted Secure Data Sanitization Platform (In Progress)

- Developed an AI-powered secure data sanitization platform that helps users identify, analyze, and permanently erase sensitive data from files, folders, and storage devices.
- Integrated Kali Linux secure wipe utilities **(shred, dd, nvme-cli)** to perform NIST SP 800-88 aligned data sanitization and prevent forensic recovery. Implemented AI-assisted **file classification, risk assessment, and trust-score** generation to help users make informed data disposal decisions.
- Building automated audit logging, PDF/JSON certificate generation, and wipe verification workflows using MongoDB and Python. Conducting forensic validation using Autopsy to verify the effectiveness of secure data destruction mechanisms.
- Technologies: Python, Flask, Electron.js, JavaScript, MongoDB, Kali Linux, SSH, Autopsy, HTML, CSS.

SafeTrack SOS – Smart Emergency Alert & Location Tracking System

- Built an IoT-based emergency alert system with real-time location tracking and SOS notifications.
- Smart Emergency Alert & Location Tracking System Developed SafeTrack SOS, an IoT-based emergency alert system that enables instant SOS triggering via ESP32 hardware button or web interface. The system captures real-time location data, generates live Google Maps links, and displays alerts on a police dashboard using interactive mapping. Designed to improve emergency response time and enhance safety during critical situations.
- Technologies: ESP32, Python, Embedded C, Google Maps API, Web Dashboard, GPS Module, GSM Module, Gyroscopic Module, LORA Module, OLED Display, TP4056, Rechargable Lithum Battery, Buzzer and Jumper Wires.

SATLINK – Cybersecure Satellite Communication Module

- Developing a secure IoT-based satellite communication system using ESP32-S3, Iridium, GNSS and BLE.
- Implementing encryption, device authentication, secure key storage and replay protection.
- Analyzing security threats across device firmware, BLE, satellite communication and cloud APIs.
- Developing secure SOS and GNSS location transmission with cloud-based emergency alerts.
- Prototype development, system integration and validation are in progress.

Graphic Design & UI/UX Projects

- Designed user interfaces, mobile app prototypes, and digital marketing materials.
- Created user-centered designs using modern UI/UX tools.
- Tools: Figma, Adobe Photoshop, Adobe Illustrator, Canva

AWARDS & ACHIEVEMENTS

- Diploma in Computer Hardware Servicing – Mohan Institute of Technology (Jul 2023)
- Graphic Designing – Fortune Innovative (Oct 2023)
- First Prize, Demo Day Project Competition – Akshaya College of Engineering and Technology, Nov 2023
- Second Prize, Logo Lingo Competition – PSG College of Arts & Science, Mar 2025
- First Prize, Paper Presentation Competition – Hindustan College of Engineering, Mar 2026
- First Prize, Tech Day Project Competition – Akshaya College of Engineering and Technology, Mar 2026
- Participant in C-DAC BOSS OS National Bug Bounty Challenge 2026 (VIT Chennai Hub)
- First Prize, Dream 2 Reality Competition – Akshaya College of Engineering and Technology, Sep 2026